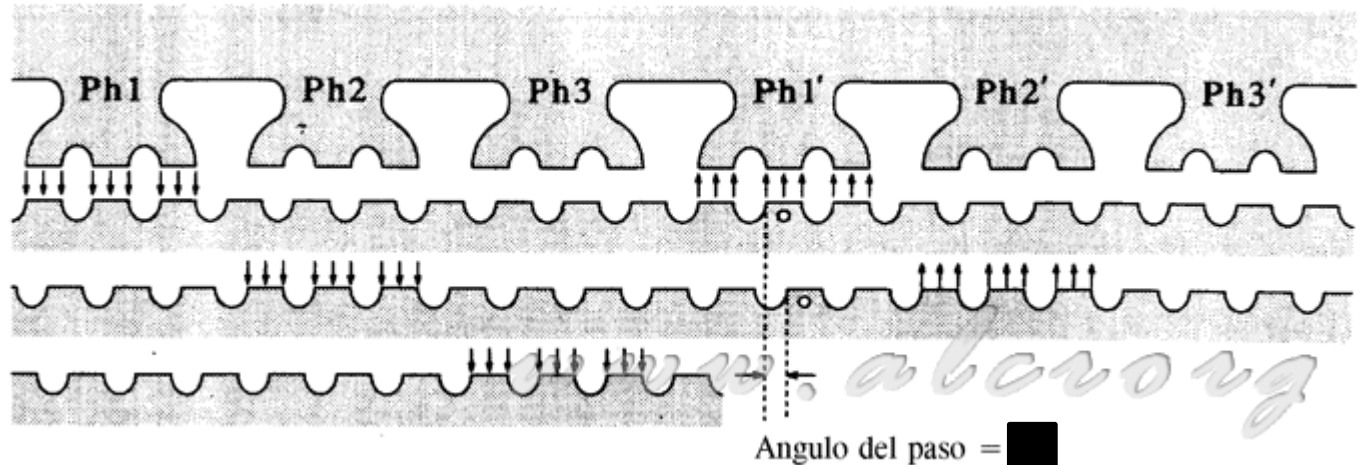
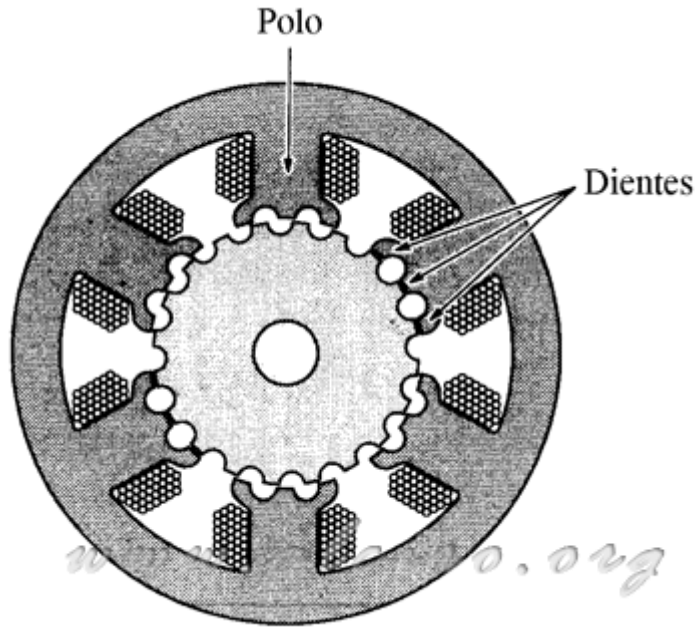
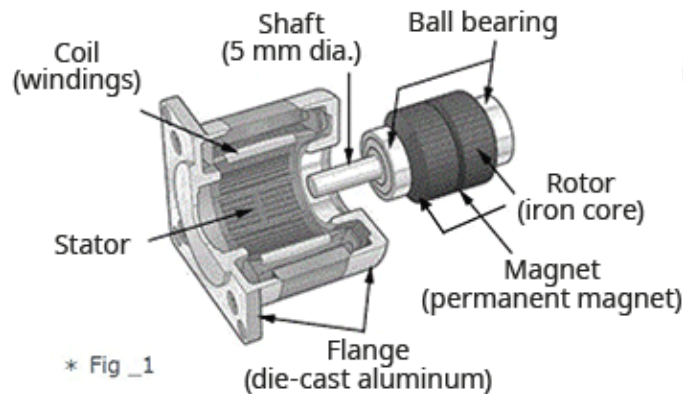
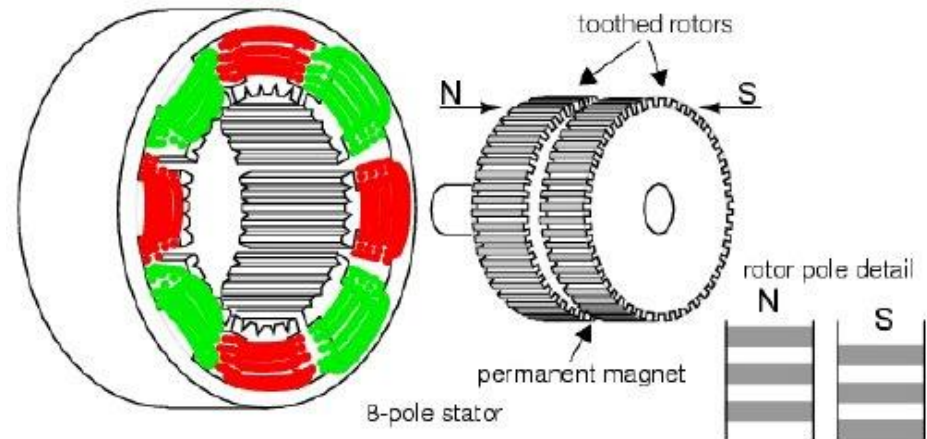
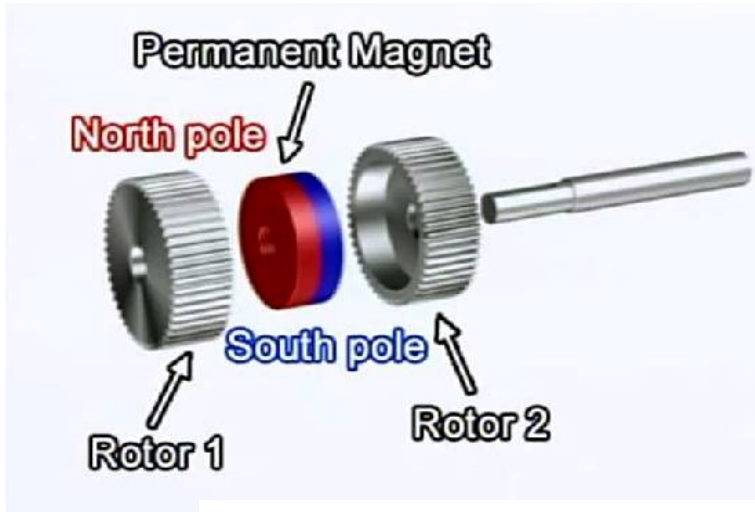


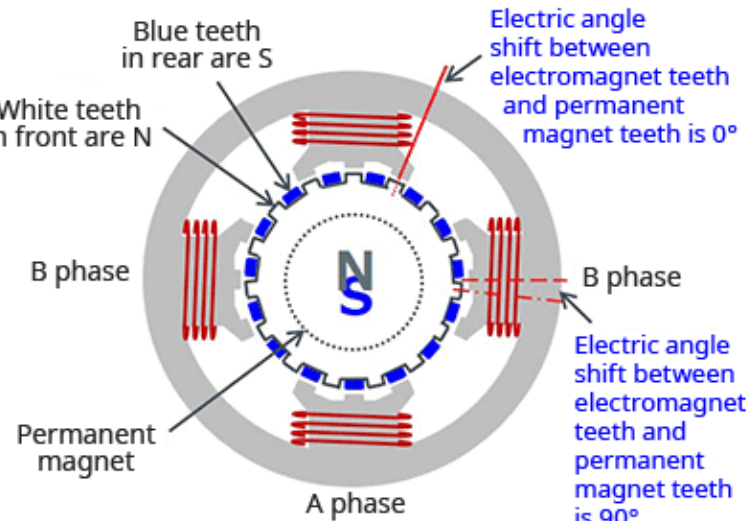
Variable Reluctance Stepper Motor



Hybrid Stepper Motor



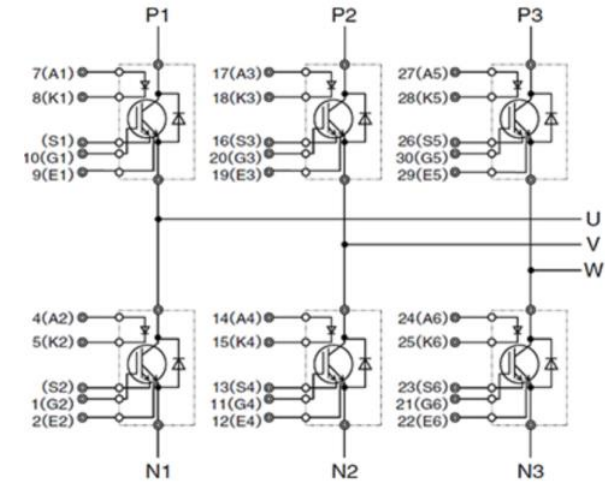
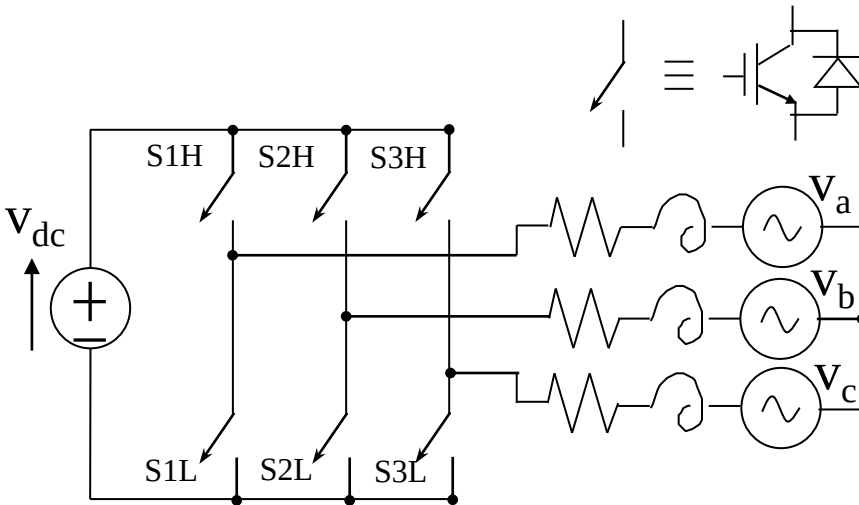
* Fig _1



Initial State

In the initial state, currents are passed such that the upper pole of the A phase (upper/lower) is S and the lower pole is N

DC/AC: three-phase voltage source inverter (VSI)



Sinusoidal PWM with third harmonic injection (inscribed circle)

$$1/\sqrt{2} V_{dc}$$

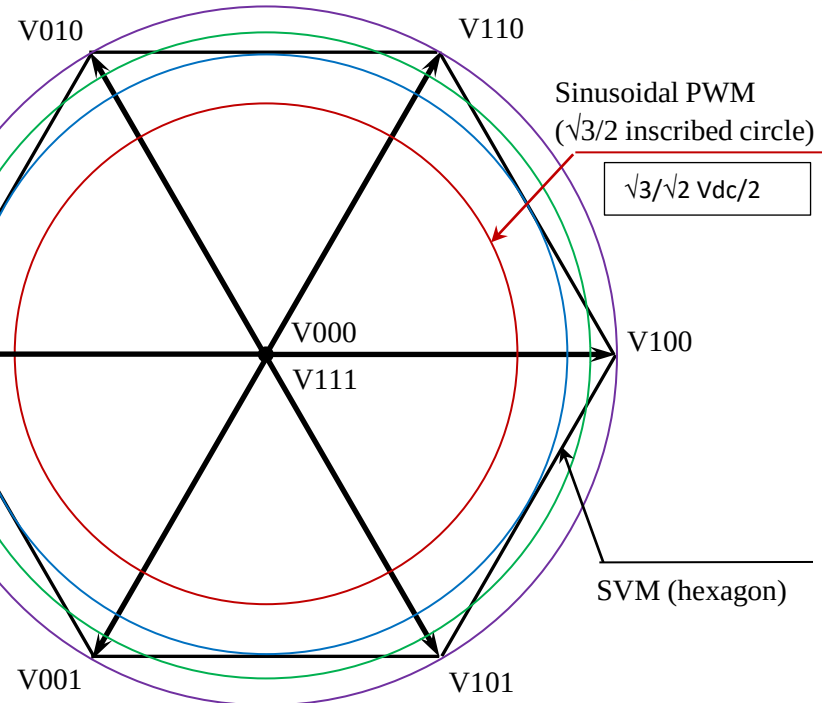
V011

Circumscribed circle

$$\sqrt{2}/\sqrt{3} V_{dc}$$

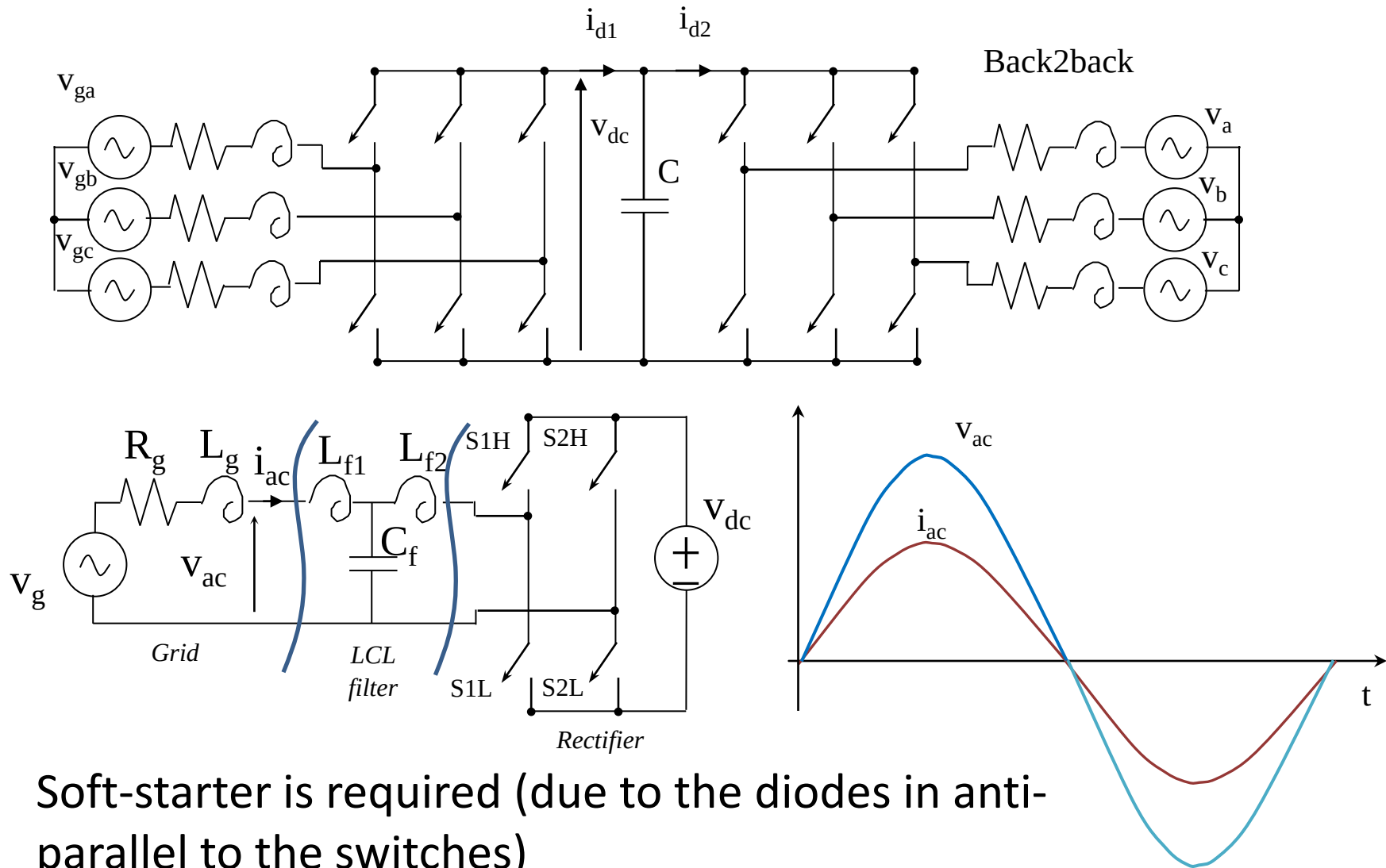
$$2/\pi \sqrt{3}/\sqrt{2} V_{dc}$$

Square wave first harmonic ($4/\pi \sqrt{3}/2$ inscribed circle)



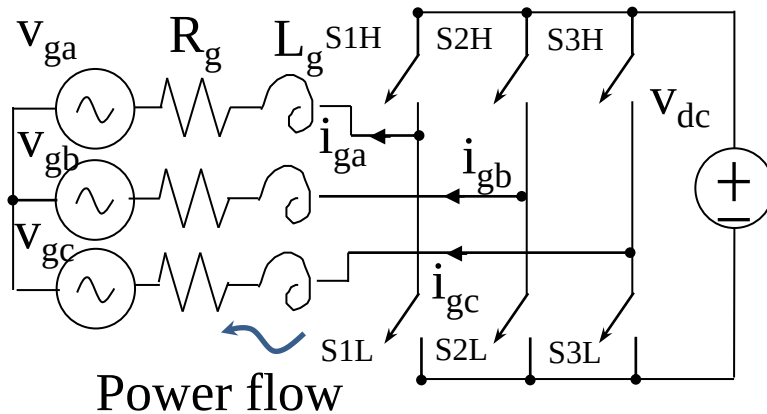
Back-to-back configuration

Called also “Regenerative rectifier” or “Active Front End” (AFE)



Soft-starter is required (due to the diodes in anti-parallel to the switches)

Regeneration



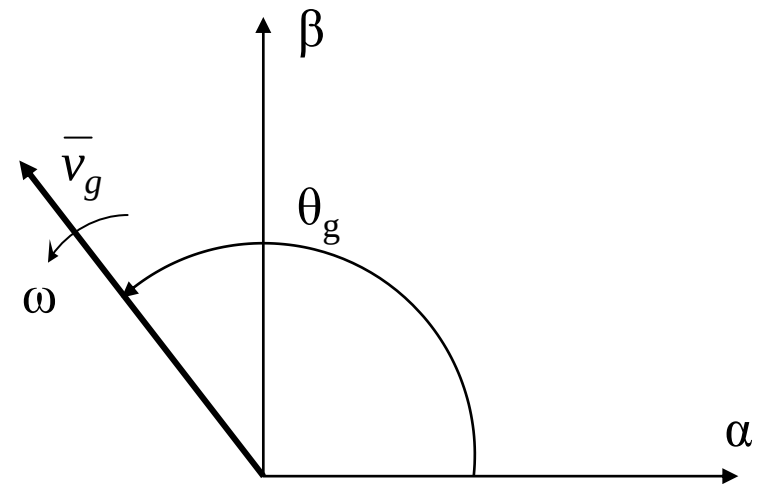
The grid has an equivalent circuit similar to that of an ac machine.

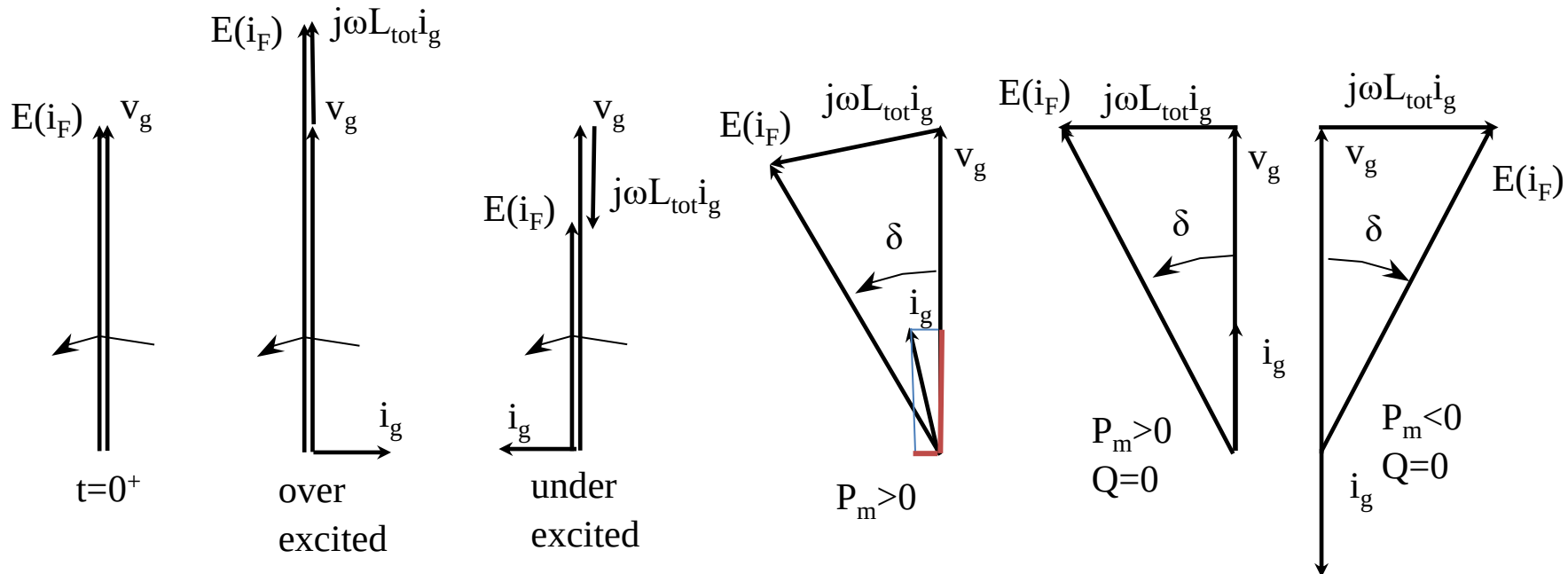
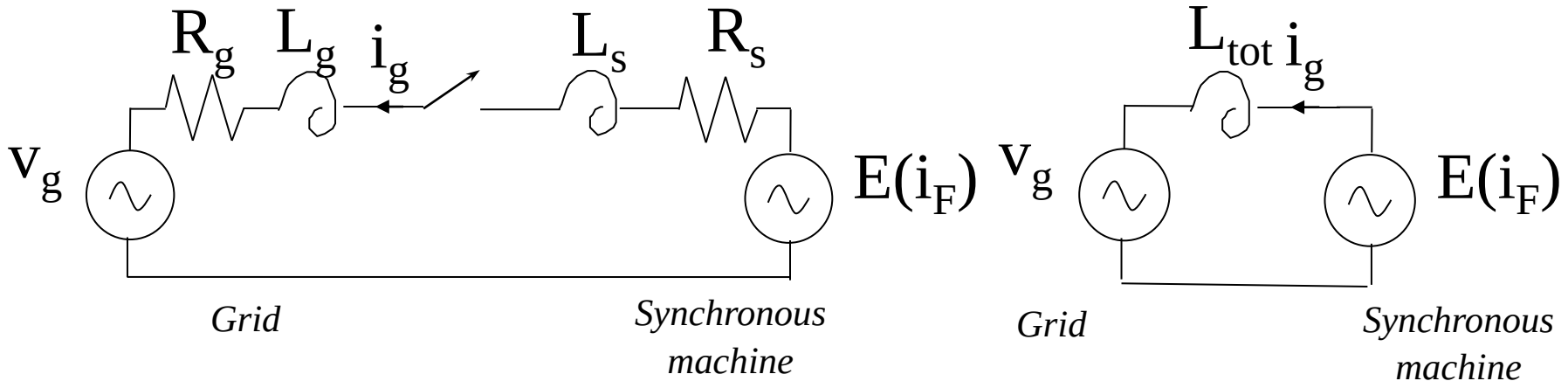
The main difference is that the voltage and the frequency are quite constant (in an ac machine the emf and the frequency are quite proportional to the mechanical speed)

The space phasor formula may be applied to the grid voltage v_g . In a stationary frame ($\alpha\beta$), it results ($v_{ga} = V_{ph\ max} \cos(\theta_g)$):

$$\overline{v_g} = V_g e^{j\theta_g}$$

Where V_g is the rms value of the line-line voltage, while θ_g the angle between the space phasor v_g and the α -axis





Control of power flow into the grid

Call v_i the voltage produced by the VSI (limited to the hexagon operating area). Choose a reference frame fixed with the grid voltage, where the imaginary axis (q) is aligned with the space phasor $\bar{v}_g$. It results:

$$\bar{v}_i = R_g \bar{i}_g + L_g p \bar{i}_g + j \dot{\theta} L_g \bar{i}_g + \bar{v}_g$$

The components along d and q axes are:

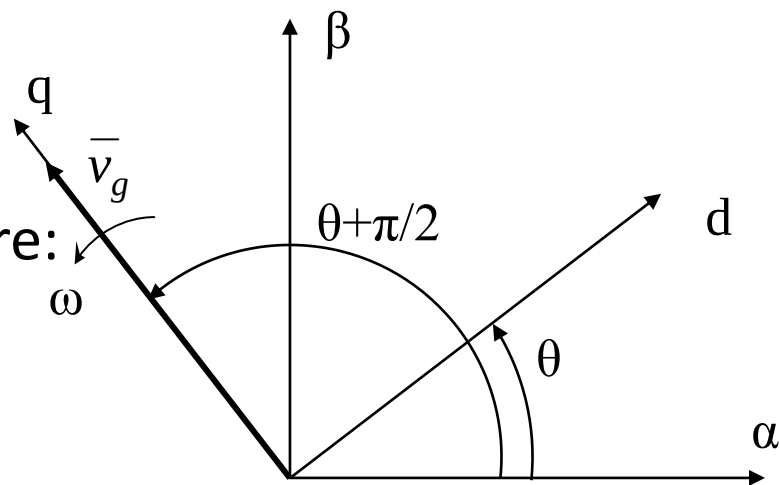
$$v_{id} = R_g i_{gd} + L_g p i_{gd} - \dot{\theta} L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + L_g p i_{gq} + \dot{\theta} L_g i_{gd} + v_g$$

The active and reactive power fed into the grid are:

$$P_g = \text{Re}(\bar{v}_g \underline{i}_g) = v_g i_{gq}$$

$$Q_g = \text{Im}(\bar{v}_g \underline{i}_g) = v_g i_{gd}$$



Control of active power fed into the grid

The derivative of the angle θ is the frequency ω .

Consider a steady state condition and a grid voltage constant in amplitude and frequency: $\overline{v_g} = V_g e^{j\omega t}$. Looking from the reference frame (dq), all the quantities are constant. Therefore:

$$\dot{\theta} = \omega$$

$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

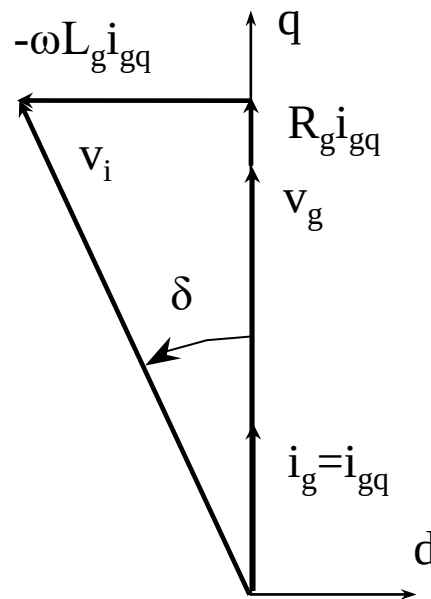
$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

If the current i_{gd} (component along the d axis) is fixed to zero, the phasor diagram is:

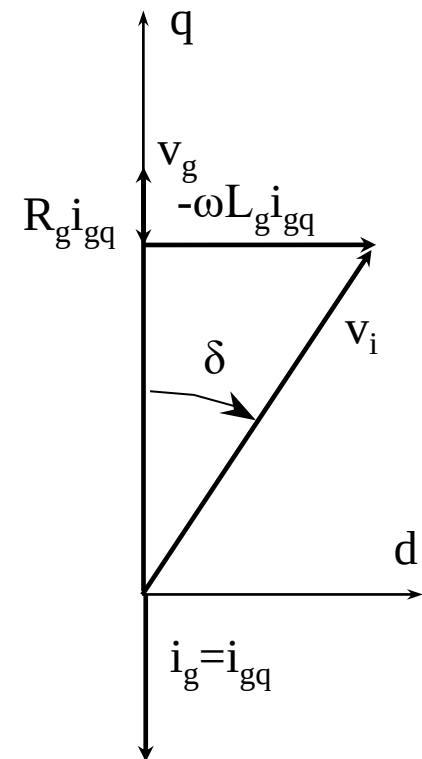
$$v_{id} = -\omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + V_g$$

Regenerative & $Q_g=0$



Rectifier & $Q_g=0$



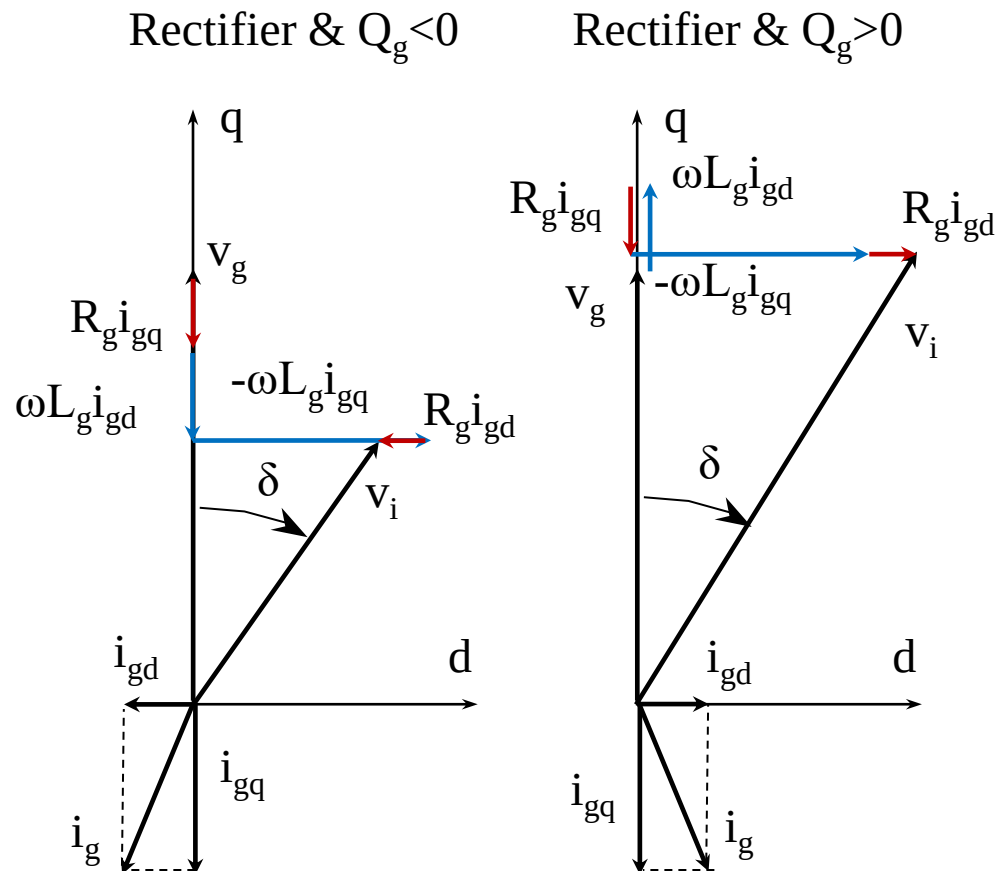
Control of active/reactive power fed into the grid

Controlling the current i_{gd} , it is possible to drive the reactive power: a positive value means that the inverter looks like a capacitor (and the grid like an inductor); vice versa for a negative value.

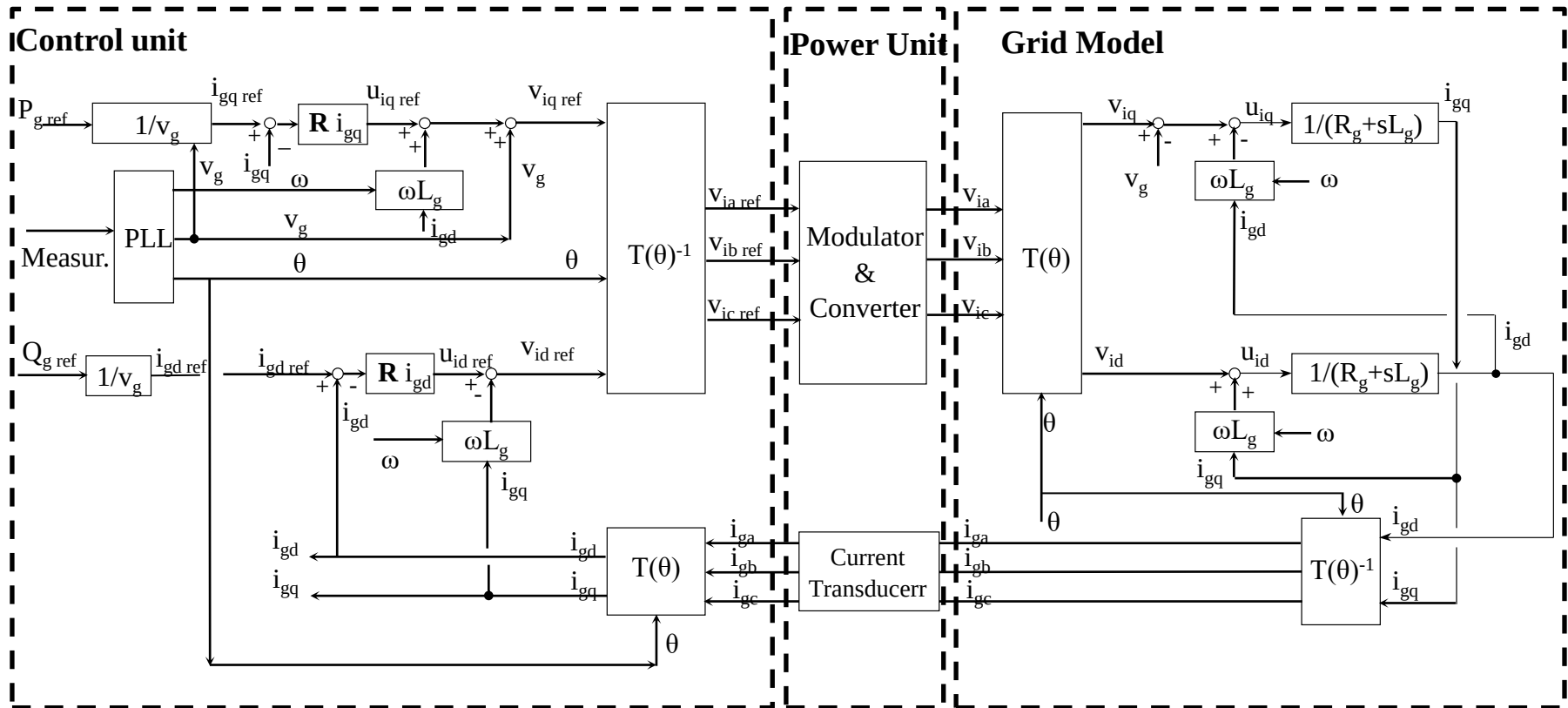
$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

The figure represents the conditions when the active power is negative (operation as rectifier). Similar approach may be done for positive value of the active power (regeneration)



Control schemes



Control schemes: comparison with ac brushless

